

Analytics Club

End-Term Work Report (2025-26)

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About the Division

Established in 2016, Analytics Club is the representative body for the interests of all students of IIT Bombay in the domain of analytics, data science and AI/ML. We carry out sessions, bootcamps and hackathons on Machine Learning, Artificial Intelligence, Big Data Handling, Data Visualization, etc. and try to promote the community of students pursuing these fields in the institute. The vision of the club also includes helping students stay informed about opportunities in the field of Analytics and similar domains by guiding them in their journey to help them build a career.

We also help the Data Analytics and Visualization team (DAV Team), who carry out analytics on real-time institute data. DAV Team consists of a small cohort of students dedicated to helping the institute with Data Analytics techniques by drawing valuable insights from the huge amount of data available at the institute level.

We also cover various topics in the interest of MTech, PhD students of the institute and their participation is very valuable to the club in all its activities. We strive to spread awareness about the plethora of opportunities available in this rapidly-growing field and help students gain relevant skills.

Internship Preparation Programme Speaker Sessions: Jun '25

With internship season on the horizon, the Analytics Club successfully organised 3 dedicated speaker sessions covering the profiles: Analytics, FMCG & Supply Chain, and Quantitative Finance.

In these sessions, students were addressed by seniors who had previously secured coveted internships at some of the most well-known firms in their respective domains. The sessions were packed with substance: covering important preparation techniques, how to handle aptitude tests and case interviews, how to strategically target companies, what resume-building for these profiles looks like, and answers to the most frequently asked questions.

Sessions by Seniors:

- **Analytics Profiles** [9th June 2025]: This session was taken by Gokul Rajaraman (Adobe), Priyanshu Gangavati (AWL Japan), Amish Sethi (Citi Bank), and Varun Ram Narayana (American Express).
- **FMCG & Supply Chain Profiles** [8th June 2025] (In collaboration with Consult Club): This session was taken by Rahul Gupta (P&G), Isha Laddha (P&G), Aditya Shrungarpure (ITC), and Monank Vijayvargiya (ITC Group).
- **Quant Profiles** [10th June 2025] (In collaboration with Finance Club): This session was taken by Deeksha Dhiwakar (Tower Research Capital), Anvesh Aggarwal (Graviton Research Capital), Nimay Shah (Optiver), and Nandnee Priya (Goldman Sachs).

Internship Preparation Bootcamp: Jun '25

Alongside the IPP speaker sessions, two comprehensive self-paced bootcamps were launched one each for Quant and Analytics profiles to give students the structured preparation they needed to crack the tests and interviews ahead of them.

The content itself was carefully curated with the help of seniors who had immense first-hand knowledge of these domains and had previously bagged internships at leading Analytics and Quant firms.

Interview Preparation Booklet: Jul '25

A comprehensive 60-page Interview Preparation Booklet was crafted, a one-stop guide for any 3rd-year student targeting an internship in the field of Analytics. What makes this booklet stand out is its authenticity: it was built entirely from in-depth interviews with 15+ seniors who had actually lived through the experience of securing analytics internships at top firms.

Placement Preparation Program (PPP): Oct '25

This program included two intensive sessions aimed at helping students prepare for placements in the Analytics and Quant domains. Senior students who had secured placements in reputed firms conducted sessions on handling interviews, test preparation, and career planning. Students were awarded incentive points only for the Quant session.

The two sessions were:

- **Analytics Profiles** [18th October '25]: Taken by Mehul Agrawal (BlackRock), Anand Narasimhan (Abacus.Ai), and Alaukik Varma (American Express). The session went deep into preparation strategies for analytics roles, the structure and format of interviews at leading firms, and how to build and communicate a strong analytical profile.
- **Quant Profiles** [1st November '25] (In collaboration with Finance Club): Taken by Swadhin Dash (Quadeye), Omm Agrawal (WorldQuant), and Ashwin Abraham (Quantbox Research). The session covered quantitative aptitude test preparation, the mathematical problem-solving techniques that matter most, and candid accounts of what interviewing at elite quant firms is really like.

Learners Space: *June '25*

Learner's Space is a learning platform where a student's enthusiasm is the only prerequisite to join a course. The Analytics Club launched two courses as part of the CS & DS Summer School for the 9th Edition of Learner's Space continuing its tradition of making high-quality, structured technical education freely accessible to all students over the summer. The courses were curated after extensive research and delivered through Notion pages updated daily with fresh course material. Each course combined exhaustive video lectures, curated reading material, graded assignments, and live online speaker sessions.

Course Name	Duration	Brief Description of the Course
Big Data Handling with Excel & SQL	4 weeks	The main aim of this course was to familiarise students with Excel and SQL, the two indispensable tools used by data-driven organisations across the world and demonstrate how they can be wielded to handle, query, and draw insights from big data. The Excel module gave students a strong fundamental understanding of the tool covering arithmetic operations, essential functions, sorting and filtering, data visualisation, and the use of pivot tables. The SQL module introduced students to the world of querying and managing big data, covering Group By and Having statements, Basic Querying, Joins, Subqueries, and best practices for building fast, secure, and scalable databases. Excel Speaker Session [18th June 2025]: Conducted by Aryan Adinath Popalghat, Senior Associate Data Scientist at Publicis Sapient, covering real-world analytical applications of Excel for data handling and analysis. SQL Speaker Session [26th June 2025]: Conducted by Raghav Rander, Growth Engineer at Kotak Life, covering scalable database design and practical querying techniques. 3 long assignments for Excel and 2 for SQL were given. Students who satisfactorily completed them received certification.

Data Science Fundamentals with Python	5 weeks	This course was designed to build a strong foundation for Data Science and Machine Learning from the ground up. Over 5 weeks, students progressed through: Basics of Python Programming; Introduction to NumPy and Pandas; Exploratory Data Analysis (EDA) and data visualisation with Matplotlib and Seaborn; Basics of Statistics and Probability; Supervised learning algorithms including Linear and Logistic Regression and Decision Trees; and Unsupervised learning including K-means and Hierarchical Clustering, PCA, and Anomaly Detection. Additional optional content for further exploration was also provided. Speaker Session [22nd June 2025]: Conducted by Amit Rohila Data Scientist at OneCard, AI Consultant at FollowFox.ai (USA), and AI Workshop Speaker & Blogger. The session dived into real-world Data Science project pipelines and demonstrated how AI is applied to solve practical problems across industries, making AI understandable to everyone. The course involved 3 weekly assignments and a final project. The criterion for certification was satisfactory performance in the final project and at least 2 out of the 3 weekly assignments.
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Personal Data Agents Hackathon : Jul '25

At a time when questions around data privacy, AI ethics, and digital ownership are more pressing than ever, the Analytics Club and the DAV Team came together to organise a first-of-its-kind hackathon on trust-native Personal Data Agents (PDAs) an emerging paradigm where individuals, not corporations, control their own data and how AI uses it.

Sponsored by Hushh AI, the hackathon ran from 22nd to 30th July 2025 and offered a prize pool of INR 1.7 Lakhs, along with internship and full-time opportunities at Hushh making it one of the most rewarding hackathons organised by the club. Participants used HushhMCP, a cutting-edge framework for building privacy-first AI applications, pushing them to think not just about what AI can do, but how it should be designed responsibly. Before the hackathon began, a dedicated launch session was conducted by Manish Sainani, CEO of Hushh where he shared industry insights and trends on decentralised data architectures, personal AI agents, and the ethics of AI.

Session on Masters in CS and DS: Aug '25

Organised a session with two distinguished speakers, both IIT Bombay Mechanical Engineering graduates (Batch of 2025), who transitioned from a non-CS background to top US universities:

- Anubha Vyasamudri : MS in Computer Science at Columbia University
- Aryan Bhosale : MS in Computer Science at University of Maryland

Their journeys highlighted how students from any engineering background can break into top CS/DS programmes. The session covered key aspects of MS applications, including programme overview (CS/AI/DS), profile building, timelines, college shortlisting, GRE/TOEFL prep, SOP writing, and securing strong LORs.

Power BI Workshop: Aug '25

A Power BI session was conducted by Arijit Chowdhury, Senior Vice President at YES Bank, covering key concepts in data visualization and dashboard creation. Participants were given an assignment, successful completion of which would lead to certification.

Stratify : Aug '25

Stratify was designed in collaboration with Consult Club to provide students with hands-on experience in business analytics through real-world datasets and problem statements. The initiative combined a structured learning programme with an industry-style project.

Participants worked across the full pipeline, including data cleaning, analysis, and deriving data-driven business recommendations, with an emphasis on structured thinking and clear communication of insights.

Key Components:

1. Self-Paced Bootcamp:

Introduced essential tools and concepts required for the projects, including Excel, Tableau, Power BI, and basic business analytics frameworks, ensuring accessibility for participants with varying experience levels.

2. Project Competition:

Four projects were conducted across business and supply chain analytics domains. Over 300 teams registered, with 100+ final submissions.

AI Newsletter: Sep '25

In a field as fast-moving as AI and machine learning, staying current is not a luxury, it is a necessity. Recognising this, the Club launched AI Pulse, a curated Instagram newsletter dedicated to keeping the IIT Bombay student community informed about the most significant developments in AI and ML, delivered in a format that is engaging and accessible.

CodeLite Analytics - Hackathon: Sep '25

The Analytics Club organised CodeLite Analytics: An AI App Hackathon in collaboration with Zoho Creator, offering a total prize pool of INR 1.1 Lakhs. The premise was both novel and timely: participants were challenged to build AI-powered applications to solve real problems faced by the institute community, using Zoho Creator's no-code platform rather than writing code from scratch.

Session on ‘Breaking into ML Research’: Oct ‘25

Organised a session to introduce IIT Bombay students to pathways in AI/ML research and make the field more accessible to undergraduates. The session was conducted by Gagan Jain (Applied Scientist at Microsoft AI; former Google DeepMind; IIT Bombay, Batch of 2022) and held online on 4th October at 8:30 PM.

Key topics covered:

- Differences between research-oriented and industry-oriented AI/ML career paths
- Approaching professors and research labs for undergraduate opportunities
- Core skills required for ML research (mathematics, programming, concepts)
- Reading and understanding research papers
- Overview of day-to-day work in AI research roles

The session provided students with a practical understanding of research pathways and actionable guidance to get started.

GenAI Workshops

A series of workshops was conducted to introduce students to practical applications of Generative AI, focusing on building, customizing, and deploying Large Language Models (LLMs) for real-world use cases. These sessions aimed to bridge the gap between theoretical understanding and hands-on implementation, equipping participants with relevant industry skills.

- **Customizing LLMs: RAG and Fine-Tuning — Oct ‘25**

The first session was conducted by Varun Raste, Data Scientist at IBM, who shared his expertise in machine learning optimization and AI-powered education technology. The workshop was hands-on, enabling students to learn and apply LLM customization techniques using Retrieval-Augmented Generation (RAG) and fine-tuning through practical, real-world use cases.

- **LangChain and LangGraph — Jan ‘25**

(Continued under the Confluence section)

Deep Learning Bootcamp: Oct ‘25

The annual Deep Learning Bootcamp returned for another edition a two-week, comprehensive, hands-on online programme covering both foundational and advanced material across two of the most exciting subfields in modern AI. With no prerequisites

required, the bootcamp was designed to be genuinely accessible: the only entry requirement was curiosity and a willingness to engage with the material seriously.

The bootcamp offered two parallel tracks, allowing students to pursue the direction that excited them most:

Computer Vision (CV) Track:

- Section 1: PyTorch and Image Classification with Neural Networks
- Section 2: Convolutional Neural Networks for Image Classification
- Section 3: Transfer Learning
- Section 4: Image Segmentation and Object Detection

Natural Language Processing (NLP) Track:

- Section 1: Natural Language Tool-Kit (NLTK) and Text Processing
- Section 2: PyTorch and Text Classification with Neural Networks
- Section 3: Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM)
- Section 4: Transfer Learning and Fine-Tuning LLMs with HuggingFace

The bootcamp also featured Basic and Advanced tracks across NLP and CV. The Basic track included 4 assignments, with any 3 required for certification, while the Advanced track involved 3 assignments and a compulsory project, with at least 2 assignments plus the project needed for certification.

WiDS 5.0 (Winter in Data Science): Dec '25 - Jan '26

The fifth edition of Winter in Data Science the Analytics Club's flagship long-form learning initiative was launched, continuing its mission of building a vibrant, practitioner-oriented community of students who learn data science by actually doing it. WiDS connects mentors with mentees through collaborative, project-based learning programmes spanning 4 to 6 weeks making it one of the most immersive and substantive learning experiences the club has to offer.

Additionally, WiDS 5.0 introduced a beginner-friendly Data Science Bootcamp, powered by School of DS, providing foundational training in core concepts and tools to help students kickstart their journey in data science before diving into projects.

WiDS 5.0 introduced two significant innovations:

- **Core Projects:** A new project category was introduced this year, focusing on the intersection of core engineering domains and Data Science. This was a deliberate move to open WiDS up to students of other engineering backgrounds enabling them to explore how data science applies directly within their own fields.
- **Official Dynamic Website:** For the first time in WiDS, a fully functional dynamic website was developed for the programme with the UGAC Web Team allowing both mentors and mentees to register, browse available projects, and manage their participation entirely through the platform. This replaced the previous Google Form-based registration workflow and gave WiDS a professional, scalable digital home commensurate with the scale and ambition of the initiative.

Convolve: *Jan '26*

The fourth edition of Convolve, pan-IIT AI/ML hackathon returned bigger, bolder, and smarter than ever before. Conducted in collaboration with 8 other IITs (IIT Guwahati, IIT Madras, IIT Delhi, IIT Kharagpur, IIT Roorkee, IIT BHU, IIT Hyderabad, and IIT Kanpur), Convolve 4.0 announced itself with a brand-new format that raised the stakes considerably: for the first time in the competition's history, Specialised Tracks were introduced, allowing participants to compete in the domain they were most passionate about.

Open to participants across the globe, the hackathon was sponsored by IDFC First Bank, Quadeye, and Qdrant. The total prize pool stood at an impressive INR 18,00,000, distributed equally across the three specialised tracks:

- GenAI Track: INR 6 Lakhs
- MAS (Multi-Agent Systems) Track: INR 6 Lakhs
- Quant Track: INR 6 Lakhs

Beyond cash prizes, the hackathon offered PPI (Pre-Placement Interview) and PPO (Pre-Placement Offer) opportunities through its industry partners. A dedicated website and Instagram account were created and maintained throughout the competition to ensure participants had a single, reliable source for all updates and announcements.

The hackathon was conducted in multiple rounds, each progressively more demanding:

- **Round 1** - Common Screening Round [10th-11th January 2026]: Registrations closed on 9th January 2026. Participants from across the globe competed in the initial screening, with teams self-selecting into their chosen specialised track.
- **Round 2** - Track-Specific Challenges [17th January 2026]: Top teams from Round 1 advanced to domain-specific problem statements tailored to their track, testing deeper technical skills and creative problem-solving in GenAI, MAS, or Quant.
- **Grand Finale**: The highest-performing teams from each track presented their solutions before a panel of expert judges from the sponsoring companies competing for both the cash prizes and coveted placement opportunities.

Orientation of Analytics Club: Jan '25

The Analytics Club conducted an orientation session to introduce freshers to the world of data science, analytics, and AI/ML. The session provided an overview of the club's vision, past initiatives, and upcoming opportunities, while highlighting how students can get involved and build skills in these domains. It aimed to spark interest, guide beginners, and help them take their first steps into the field of analytics.

Analytics Career Guidebook: Jan '25

Developed a comprehensive guidebook to help students navigate diverse analytics career paths across various domains. The guidebook outlined key roles (data analyst, ML engineer, quant researcher, business analyst, supply chain analyst), detailing responsibilities, required skills, hiring domains, and preparation strategies. It serves as a structured, one-stop resource for students to understand and pursue analytics careers.

Confluence : Jan '25

Confluence 2026 was the second edition of our flagship fest organised in collaboration with the Consult Club.

It contained keynote speaker sessions, workshops, and fun, interactive games. This 3-day fest had an amalgamation of every component required to introduce and immerse students into the world of analytics and consult.

- **Day 1**

Keynote Speaker Session by Mr. Sairam Prasad, CEO of Suzlon

We had the privilege of hosting an insightful Keynote Speaker Session with Mr. Sairam Prasad, where he shared his expertise on leadership in the digital era,

providing key insights on aspects that remain timeless, aspects that have drastically changed, and technology impacts the strategies of companies. He also discussed his predictions of the impact of the new industry wave - AI - on leadership. It was conducted online.

StratX - The Startup Trivia Game

As part of our fest, we organized a highly engaging and interactive startup trivia game, called StratX. The game encouraged participants to apply logical reasoning, balanced thinking, and quick judgement to make real time strategy decisions for their emerging company. With an enthusiastic turnout, the event fostered lively discussions, making it both an intellectually stimulating and fun-filled experience. The dynamic format kept the energy high, ensuring active participation and a competitive yet collaborative atmosphere.

Da Vinci Code – The Ultimate Crypt Hunt Challenge

The second edition of the online cryptic hunt titled 'Da Vinci Code' was launched alongside Confluence. This event was designed to challenge participants' problem-solving abilities, creativity, and analytical thinking. It featured a series of hidden clues, intricate riddles, and complex puzzles that required participants to think critically and strategically to progress through each stage.

- **Day 2**

Public Policy Keynote Session

The public policy keynote session was spearheaded by IAS Vimala R., who provided invaluable insights into the world of policy making and government functionality. The session explored her personal journey as a bureaucrat- the challenges she faced, achievements she secured. She discussed the skillset required to enter the world of public policy, the motivation and passion needed, and concluded the session with a briefing on the UPSC process.

CaseFront Finals

The finals of the CaseFront case competition were conducted offline, judged by an executive sent from Masters' Union. Finalist teams had to present their case decks live, and were quizzed on their knowledge and strategy. Judged on clarity of thought, innovation, and feasibility of solution, the finals were engaging and competitive, and were executed smoothly.

The Outlier – Murder Mystery Game

Conducted a gamified murder mystery event integrating data analytics and critical thinking. Participants, in teams, solved a storyline-driven case through clues and twists, applying logic and teamwork. The event incorporated hands-on analysis using Google Colab, Python, Pandas, and data visualization tools to interpret clues and derive insights. Delivered a fun, engaging learning experience while reinforcing practical analytics skills.

- **Day 3**

Workshop on LangChain and LangGraph:

A comprehensive and practical workshop on LangChain and LangGraph was conducted. The session was led by Tarun Jain, Google developer expert in AI. He has extensive industry and teaching experience, including previous roles at AI planet. The workshop gave participants an in-depth understanding of Langchains, their practical applications, and the integration of their functionality into real-life scenarios. A live demonstration of LangChain and LangGraph was conducted, with a strong focus on building out AI systems from scratch on Python.

Keynote Speaker Session : ‘Converging AI, Analytics, Big Data, and Automation’

Conducted by Vishal and Nitinn Agarwal, founders of Locobuzz, this session explored key concepts such as AI, Big Data, Automation, and Machine Learning, and how they interrelate and shape the future. Drawing from their experience building a data analytics company, the speakers shared insights on data collection, analytics, and industry trends. They also addressed ethical considerations and emerging regulatory guardrails around AI and public data use. The session was informal and interactive, with knowledge shared through an engaging Q&A with students.

Infographics

Analytics Club continued its commitment to keeping data science and AI alive as everyday conversations on campus through a regular series of educational infographics published on Instagram. Each infographic is a carefully researched, ten-slide deep dive into a topic that sits at the intersection of data science and the world students actually inhabit. This year's infographics covered:

- **AI Bosses in Games:** An exploration of how intelligent game opponents are built using AI from rule-based systems to reinforcement learning and how the field of game design has become an unlikely testbed for some of the most interesting AI research happening today. Where game design meets artificial intelligence.
- **AI Tools and Hacks:** A curated collection of the most powerful free and freemium AI tools available today compiled specifically to help students supercharge their productivity and explore what modern AI can do for them in daily academic and professional life. Because still using only ChatGPT means barely scratching the surface.
- **Why Do You Get Targeted Ads After Just Talking?:** A sobering and eye-opening look at how apps collect and use personal data, how targeted advertising systems work under the hood, and concrete measures students can take to protect their privacy in an increasingly surveillance-heavy digital world. Is your phone listening to you? Swipe to find out.
- **Why Do We Procrastinate Before Exams?:** Released during midterm season, this infographic combined behavioural science with AI-informed strategies for exam preparation helping students understand the psychological roots of procrastination and offering practical, research-backed techniques to break through it when it matters most.